

Exploring Proxemics in Volumetric Video vs. Face-to-Face through Charades

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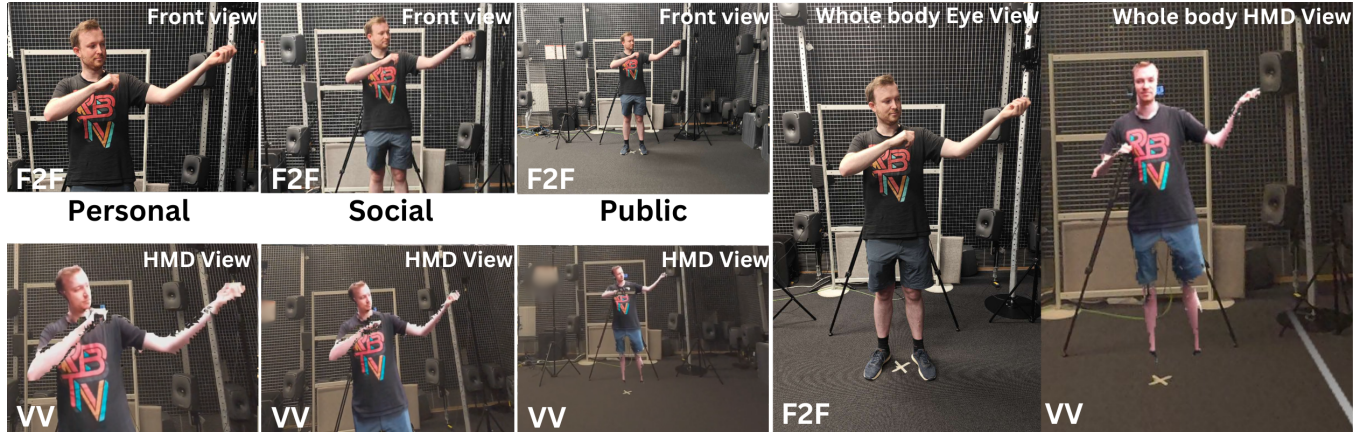


Figure 1: Exploring Proxemics in Volumetric Video vs. Face-to-Face through Charades

Abstract

We aim at exploring how people navigate different interpersonal distances during a charades game and how that affects the effectiveness of charades guessing. We conducted a study ($N = 54$) using a split-plot design. In each trial, one participant acted as the actor, while another served as the guesser, positioned at personal, social, or public distances (between-participants factor) from their partner. The study featured two conditions: face-to-face (F2F) and augmented reality (AR) (within-participant factor), with participants wearing an Oculus Quest 3 to view the volumetric video of their game partner in the AR condition. Our results show that participants tend to remain within personal and social distances, with a more exploratory movement when watching the volumetric video of their partner. Also, participants tended to move towards the social distance when starting at the public distance. Finally,

guessing effectiveness was higher in the F2F condition and at the social distance.

CCS Concepts

• **Human-centered computing** → **Mixed / augmented reality**; *Usability testing*.

Keywords

Augmented Reality, Usability Testing, Volumetric Video, Charades, Proxemics

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1 Introduction

Gesture understanding is a critical component of human communication, and it may be difficult to transmit in immersive environments. Augmented Reality (AR) and Face-to-Face (F2F) interactions differ significantly in non-verbal communication, particularly in transmitting and recognizing gestures [13, 33, 36]. While facial expressions and subtle body movements are critical for communication, AR systems often do not effectively capture them due to occlusion from Head-Mounted Displays (HMDs), which can hinder eye contact and facial expression interpretation [33]. Fine hand movements also remain challenging to detect due to current tracking limitations [26]. Another aspect that can affect nonverbal cues

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interpretation is latency, as AR systems require processing time, impacting real-time responsiveness [13]. These factors can affect interpreting non-verbal forms of behavior, especially in immersive environments.

An important form of nonverbal behavior is how people move in the space in relation to another person, i.e., navigating proxemic behavior. Proxemics serves as a theoretical framework for understanding how users establish and maintain comfortable spatial experiences [15]. These nonverbal behaviors account for approximately 60% of human interaction [7, 29] and define the dynamics of social interactions [25]. While physical-world proxemic patterns are well-documented, their application and adaptation in Augmented Reality (AR) environments present unique challenges, particularly, for virtual humans. Understanding how a person located in the physical world perceives and responds to different proxemic zones when engaging with volumetric representations of another person remains a critical research challenge.

Research on Volumetric Avatars (VA) has focused primarily on verbal communication and visual quality [2, 10, 30, 31, 36, 40], with only a few efforts considering interpersonal distance dynamics. Another aspect is the accuracy of distance perception, which can vary between devices and media. Leoste et al. [24] showed that users maintain similar interpersonal distances with telepresence robots and AR avatars as they do in face-to-face interactions. However, Novick and Rodriguez [28] observed that people tend to position themselves closer to virtual agents in AR compared to real-world F2F interactions. In addition, the type of device used plays a role in distance estimation, for instance, smartphones often produce accurate distance estimates [6], whereas tablets show distortions, with observers expanding midpoint intervals at 15 m but compressing them at 30 m [35]. These mixed findings underscore the need for deeper investigation into proxemic behavior and spatial interaction patterns in AR.

In this paper, we examine how proxemic behavior influence gesture understanding and interpretation across different visual representations of a communication partner. For that, we carried out a study using a mixed methods approach, combining within-subject comparisons, F2F vs. AR using volumetric videos (VV), and between-subject factors based on Hall's proxemic zones (public, social, and personal distances) [15]. Participants performed a charades game task in which they guessed different words based only on the gestures/acting of their game partner. Here, we analyze how participants move in these different spatial zones and the effectiveness in guessing accurately the word that was acted out.

This research is of an exploratory nature and shaped by the following Research Questions (RQs):

- RQ1: How do people navigate interpersonal distances (personal, social, public) when identifying gestures in F2F compared to VV?
- RQ2: How does the effectiveness of recognizing gestures at different interpersonal distances (personal, social, public) compare in F2F to VV?

2 Related work

2.1 Proxemics in F2F and in digital spaces

A key element of nonverbal communication is how close or distant people stand with respect to another person. These distances are often unconscious and influence social dynamics. Hall [15] categorized human interaction zones into four concentric ranges around the person: intimate (< 0.45 m), personal (0.45 m - 1.2 m), social (1.2 m - 3.6 m), and public (> 3.6 m). These zones reflect the varying degrees of emotional closeness or formality in interactions and are influenced by cultural norms and social contexts.

Studies investigating interpersonal distance in F2F communication show that proxemic behavior depends on context, personal characteristics, and social relationships [16]. Regarding context, cultural aspects were found to influence spatial arrangements, with members from northern societies found to maintain greater distances than members from southern cultural backgrounds, where closer physical proximity is more acceptable [18]. Further factors are related to the setting, namely lighting conditions [1], number of persons and temperature [14], and dimensions of the space [37]. In terms of personal characteristics, gender differences have been observed where women are more likely to sit closer to another woman than to men [32, 34]. Huang et al. [19] found that personal spaces between a person in the physical world and an AR agent are modulated by gender, similar to F2F. With regard to social relationships, it has been found that people stand closer if their relation is more intimate [8].

In VR, using a polygon-based 3D avatar, Hecht et al. [17] found that the personal space is circular and has a radius of approximately 1 m in F2F and VR. The emotion displayed on the face can also influence interpersonal distance [5]. For instance, angry expressions led to people keeping a larger distance [5], which similar to F2F behaviors. Besides, participants maintained greater distance from virtual humans when approaching their fronts compared to their backs [3]. Thus, the direction of approaching an individual influence interpersonal distance [20]. Additionally, there are other technical factors that can affect spatial dynamics in digital spaces, such as the field of view that can influence where a person stands in relation to a volumetric avatars [41].

In mediated environments, understanding how people move in the physical space, considering a virtual representation of another person or multiple persons remains crucial to determine the factors that influence this movement.

3 Experimental Design

The experimental design employs a mixed-methods approach to evaluate interpersonal distance dynamics. Our within-participants factor is the environment type (F2F and VV, with VV realized in AR) while the between-participants factor is the spatial arrangement. A total of 54 participants took part in the study. To control for factors based on Hall's theory [15], participants were explicitly assigned to groups in which they were paired only with strangers. This design ensured that all interactions took place between unfamiliar individuals, each experiencing both conditions, once in VV and once in F2F. We considered Hall's classification [15] and used three distances: personal distance $0.5 - 1$ m, social distance $1 - 4$ m, and public distance over 4 m. Here, we set initial positions within each

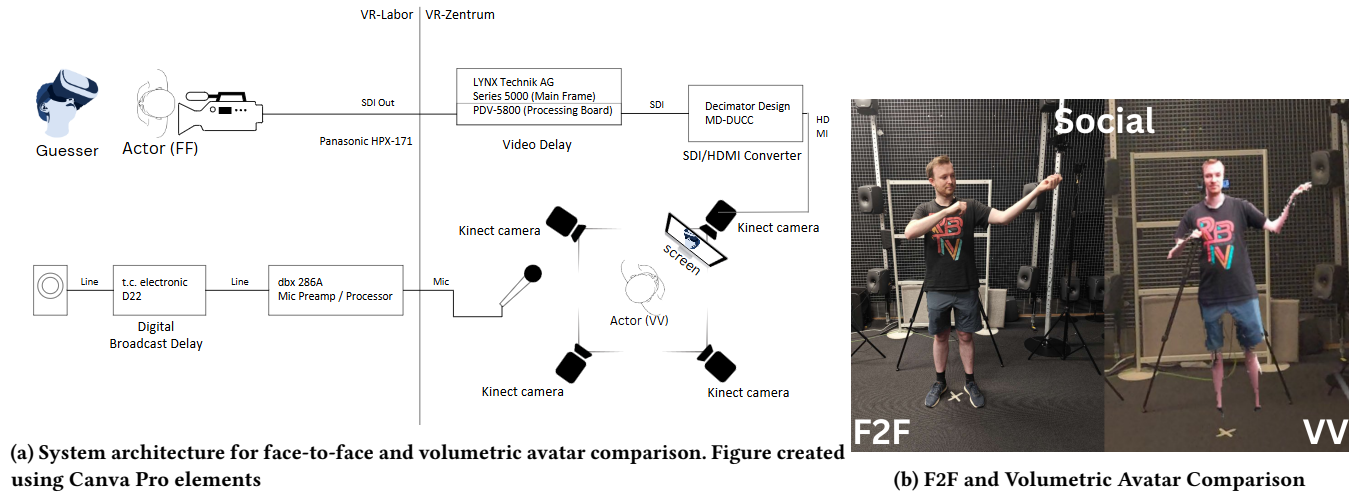


Figure 2: Comparison between F2F and Volumetric Avatar systems.

distance, i.e., personal 0.75 m, social 2.5 m, public 4 m. Each of these original positions was assigned to 18 participants. From these initial positions, participants were permitted to move freely within a circle with a radius of 1 m centered on their starting position. As dependent variables, we collected information on how people moved in the space and the effectiveness of guessing the words acted out, as a measure of communication effectiveness. To collect data on how participants' moved in the space, we tracked participants' head movements using the information from the HMD (in VV) and HTC Vive Trackers (in F2F, mounted on participant's head). Communication effectiveness was assessed based on the number of charades guessed correctly.

3.1 Task and Procedure

We used a charades game similar to the one used in [39] in which participants can either take on the role of a guesser or of an actor. The actor communicates a given word through hand gestures, arm movements, full-body movements, and facial expressions. Participants played in two rounds where they switched roles. We used a pool of 24 charade words from Wu et al.'s study [39], including actions, verbs and objects such as: pillow, drum, basketball, piano, guitar, and hunger. Each participant guessed six different words per round, with each round lasting three minutes. Participants were informed of this in advance. The order of word presentations was randomized.

The procedure of the experiment can be described as follows: first, participants are briefed on the study's goal; then, they read and sign an informed consent sheet; next, we proceed to introduce the equipment and the charades game dynamics; after that, participants start with the experimental task. In the VV condition, the actor performs gestures in a separate room and is projected to the guesser via a live volumetric video feed (see Figure 2a). The guesser, equipped with an AR headset, attempts to identify the charade using only the actor's non-verbal cues. In the F2F condition, both the guesser and the actor are in the same room, interacting directly

without the use of an AR headset. Finally, after task completion, participants filled out a compensation form.

3.2 Participants

Fifty seven participants took part in this study; however, data from 3 participants were excluded due to technical issues. Among the remaining participants, 31 self-identified as male and 23 as female. Ages ranged from 19 to 45 years ($M = 26.59$, $SD = 4.24$). Among the participants, 17 reported using visual aids, primarily glasses, while 37 reported not using any visual aids. In terms of AR/VR experience, 35 participants indicated familiarity with AR, VR or both. Participation was voluntary and participants received a compensation of 10 euros. The study adhered to ethical guidelines, including the Declaration of Helsinki, and was approved by the ethics committee of Technische Universität Ilmenau to ensure proper conduct and respect for participant rights. No specific measures were implemented to ensure sample diversity.

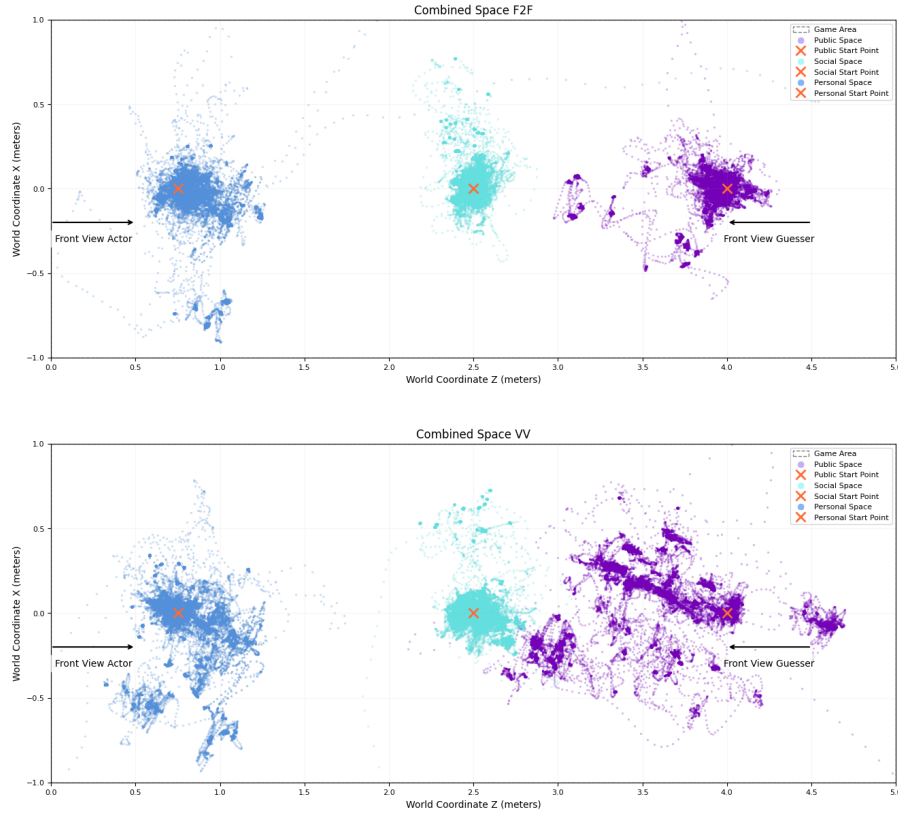
3.3 Apparatus and Experimental Setup

In our experiment, we used an Oculus Quest 3 headset in video see-through mode, enabling the guesser to observe the actor's gestures through volumetric representation, see Figure 2a. Our setup consists of two separate rooms configured to compare F2F and volumetric avatar interactions. In the F2F condition, both actor and guesser were in the same room, while for VV they were in different rooms.

3.3.1 Hardware and software Configuration. The experimental setup consists of three main components: a capture system, a video processing chain, and an audio system. The capture system employs four Kinect cameras positioned around the actor to capture volumetric data in real-time, ensuring comprehensive coverage of all movements and gestures. For video processing, the setup utilizes a Panasonic HPX-171 camera for video capture for the actor to see the guesser as a "traditional" 2D video, combined with a LYNX Technik AG Series 5000 featuring a PDV-5800 processing board for audiovisual delay management. As the volumetric avatar

Table 1: Contingency Matrix for Movement Patterns Across Proxemic Distances

	Distance Type	VV move: Likely	VV move: Unlikely	P value
F2F move: Likely	Personal	1	2	0.400
F2F move: Unlikely		3	12	
F2F move: Likely	Social	0	0	0.248
F2F move: Unlikely		3	15	
F2F move: Likely	Public	10	0	0.013
F2F move: Unlikely		8	0	

**Figure 3: Top View Position Tracking Comparison Between F2F and VV Environments Across Proxemic Distances.**

system used has around 1 s delay, this way the audio system incorporates a Digital Broadcast Delay (t.c. electronic D22) of also 1 s, to match the volumetric avatar, along with the dbx 286A microphone preamp/processor for audio processing. The 2D screen setup ensures consistent gesture visibility without requiring physical movement from the actor, maintaining experimental control, see Figure 2a. In terms of software, we used a modular application names VR4MORE to integrate volumetric video content from Consensive[12]. This system processes the data from the Kinect cameras to create real-time volumetric avatars, which are then rendered through the Quest 3's video see-through display.

4 Results

To better visualize the movement pattern of the guesser, we created scatter plots that include all guesser's movements, as shown in Fig.3. To evaluate how participants moved and determine movement patterns, we analyzed the collected head movement data with two variables: one is the duration criterion, and the other is the z-axis (forward/backward direction displacement).i.e., maintaining displacement over the duration criterion.

We set a duration criterion of 10 seconds in our analysis. This is supported by Fujimoto et al. [11], who observed that head displacement responses to AR content were most pronounced within the first 10–15 seconds of their 30-second tests. In addition, Lee et al. [23] mentioned that healthy standing participants typically exhibit

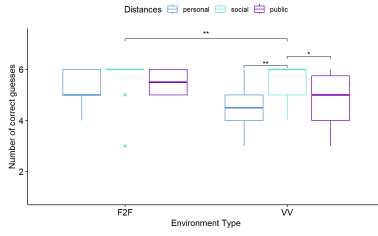


Figure 4: Communication Effectiveness (charades guessed correctly)

a diameter of head displacements (anterior-posterior) ranging from 49.12 ± 8.92 cm (without AR content) to 51.69 ± 12.68 cm (with AR content).

Thus, when our data showed head displacements along the z -axis ($|z| \geq 25$ cm) that were maintained for at least 10 seconds, where the z -axis represents the forward/backward direction, we classified this as walking (i.e., foot movement). In other words, if a participant moved more than 25 cm from their original position and sustained that displacement for at least 10 seconds, it was considered indicative of walking rather than mere postural adjustment.

We used the McNemar test to analyze the likelihood of participants to move outside of their initial position, e.g., “*move: likely*” was defined as forward/backward displacement beyond the normalized threshold ($|z| \geq 25$ cm), while “*move: unlikely*” referred to forward/backward displacements within this range, reflecting the absence of movement. These movement tendencies were compared between F2F and VV conditions, forming the paired dataset analyzed in Table 1.

To evaluate communication effectiveness (number of charades guessed correctly), we used Aligned Rank Transform (ART) [21] as the normality assumption was not met, assessed by using QQ-plots for visual inspection together with the Shapiro-Wilk test. When there was a significant difference, we used ART-C with Bonferroni correction.

4.1 Movement Patterns Analysis

A summary of the results can be seen in Table 1 and Fig. 3, aiming to illustrate and analyze how participants moved throughout the entire experiment. The figure 3 includes all movements.

In Fig. 3, we can observe that for *personal distance* in the VV environment (left), the data points exhibit more dispersion than for F2F (right), indicating slightly greater exploratory movement. At *social distances*, in both VV (left) and F2F (right) environments, we observe a dense central cluster of points, indicating a strong tendency for guessers to maintain their positions. The VV environment shows slightly more dispersed points around the social distance range.

For personal ($p = 1$) and social distances ($p = .248$), the McNemar test did not produce statistically significant differences, suggesting similar movement tendencies between F2F and VV in both personal and social distances.

For public distances, a statistically significant difference was observed ($p = .013$), highlighting distinct movement behaviors

between F2F and VV conditions at this range. The VV environment shows more widespread and diverse movement patterns, with points distributed across a broader range of coordinates and multiple distinct clusters forming at various distances. In contrast, the F2F environment displays more structured movement patterns, with clearer directional trends and more concentrated clusters, particularly showing movement toward closer interaction distances.

4.2 Communication Effectiveness

We did not find a two-way interaction effect of distances and environment type ($F(2,51) = 2.905$, $p = .06$, $\eta_p^2 = .10$). However, our results show a main effect of distances ($F(1,51) = 11.096$, $p = .001$, $\eta_p^2 = .18$) and environment type ($F(2,51) = 13.241$, $p < .001$, $\eta_p^2 = .34$) individually, see Fig. 4. Regarding distances, post-hoc pairwise comparisons show significantly higher effectiveness at the social distance ($M = 5.67$, $SD = .68$) compared to the personal ($M = 4.94$, $SD = .86$, $p < .001$) or public distances ($M = 5.17$, $SD = .85$, $p < .003$). For environment type, the F2F condition led to a significantly higher number of charades guessed correctly ($M = 5.52$, $SD = .64$) compared to VV ($M = 5$, $SD = .95$).

5 Discussion

The following paragraphs put our results in perspective to our RQs.

5.1 How do people navigate interpersonal distances (personal, social, public) when identifying gestures in F2F compared to VV?

Our findings reveal similar movement behaviors in F2F and VV, with differences at the public distance. Interestingly, within personal distance, the scatter plots reveal that movement tends to be more exploratory in VV compared to F2F.

In AR with VV avatars, personal distance remains guided by ingrained cultural and social norms, however with differences from F2F. The similarity complies to what is proposed in Hall’s proxemics model [15] and supported by other communication studies [4, 7]. This can be seen in the movement patterns where participants, all unfamiliar with their partners, tended to move away from them, especially during F2F. In VV, however, participants exhibited dispersed movement not only backward but also forward and to the sides. Iachini et al. [20] compared comfort distances in VR and found that 70 cm is a comfortable distance between a person and a virtual human. This tendency is also visible in our movement pattern analysis for AR, where some participants moved even closer from the initial position.

In terms of public distance, our results showed a significant difference of movement pattern between F2F and VV. Fig. 3 shows that participants in both conditions tended to move forward, especially during the VV condition, where participants shift to the social distance and a widespread distribution of movement can be seen. The shift towards the social distance, namely, towards the actor, can likely be explained by the participants’ desire to improve their visibility of gestures at the closer distance.

The wider spread of movement during VV can be related to findings mentioning that VV enables users to navigate spaces more freely, observe objects from multiple angles, and track dynamic

elements [41]. Additionally, there is a tendency to move closer for a better visual perception of VV [22].

5.2 How does the effectiveness of recognizing gestures at different interpersonal distances (personal, social, public) compare in F2F to VV?

When comparing gesture interpretation accuracy in F2F to VV, our results show that F2F outperforms VV.

These limitations stem from technical constraints in rendering volumetric video (VV). Participants observed that certain body parts—such as feet, fingers, and the top of the head—were sometimes partially visible or missing, affecting the accuracy of our charades-based study on nonverbal communication and gesture recognition. However, this also highlights the test’s feasibility, as comparing VV with F2F interactions allows us to quantify the system’s usability and its impact on gesture-based communication. Moreover, our findings highlights a promising direction for improvement: prioritizing hand and face tracking in volumetric video reconstruction. Enhancing these elements could yield more perceptually complete and effective representations, especially in tasks where facial expressions and hand gestures are critical. Similar efforts have been explored by Nehvi and Dong et al. in their work on volumetric capture and reconstruction [9, 27].

Regarding interpreting gestures at different distances, our results show that participants could guess the highest number of charades at the social distance compared to personal or public distances. We attribute a lower effectiveness of gesture interpretation at the public distance to the reduced visibility that a large distance involves. Additionally, during the VV condition, the reduced visibility of body parts mentioned above may have influenced these results. This can be seen also in Fig. 3 where participants show a tendency to move forward in both F2F and VV. Regarding the personal distance results, we did not expect lowered effectiveness as we considered that a closer distance would allow participants to clearly see and recognize gestures. Here, we set 0.75 m as the initial position, for personal space and we argue that this distance may have been considered too close and led to discomfort between unfamiliar participants, leading to giving up on their guesses, hence possibly explaining the reduced effectiveness. Fig. 3 also shows that participants tended to move backwards from their initial position, which is particularly prominent in the F2F condition.

We call for a note of caution on these results, as other factors, such as the acting abilities and culture of the actor may have influenced gesture interpretation effectiveness. For instance, we observed that the charades to represent a flute were enacted differently when the actor was of Asian descent. Also, as the interpersonal distance was a between-participant factor, such results could be due to the specific set-up of the individual, per-distance pairs.

6 Limitations

One technical limitation lies in the modular application used to integrate VV from VR4MORE application (in AR mode) from Consensus [12]. The quality of the VV may have impacted the participants’ ability to fully recognize, engage with or respond naturally to the virtual representation of their communication partners. Also,

we used the Quest 3’s video see-through mode which differs from natural vision and may affect depth perception, potentially altering the perception of spatial relationships and, consequently, their proxemic behavior. Here, in a future study we plan to use the see-through mode and wearing of an HMD also in the F2F condition.

Another limitation lies in the use of $|z| \geq 25\text{cm}$ as a threshold to define movement beyond personal distance in the more detailed statistical analysis. This criterion might have excluded smaller, nuanced movements, particularly in this dynamic social contexts where movement patterns often display subtleties not captured by our current movement analysis.

Our results are limited by the demographics of our participants, who were predominantly young adults. This age group tends to exhibit significantly closer distancing behaviors with peers compared to older individuals, thereby reducing variability in proxemic adjustments [38]. Similarly, in Iachini et al.’s study [20], the preferred interpersonal distance of 70 cm between a person and a virtual human was based on young students aged 18–37 years ($M = 22.3$, $SD = 4.4$). This may indirectly explain why our participants were more likely to remain within personal space.

7 Conclusions and Future work

This proxemic study of user behavior, through the integration of charades tests, highlights the nuanced interplay between proxemic distances and non-verbal communication modes, pointing to certain behavioral patterns in volumetric video and face-to-face interactions.

Firstly, for interpersonal distance navigation, participants generally remained stationary within the social and personal distance zones in both VV and F2F environments. However, VV encouraged or required more exploratory movement, particularly at public distances, where significant differences emerged.

Secondly, our results of charades accuracy indicate that the technical limitations of VV, i.e., visibility of key body parts, such as feet, fingers, and the top of the head hinder visibility and interpretation of gestures. To be specific, gesture interpretation was most effective at social distances, while reduced visibility at public distances and potential discomfort at personal distances influenced performance.

For future exploration, one key question is whether improving avatar rendering quality could lead to more natural user behavior and increased charades identification performance for VV. Additionally, comparing video see-through and optical see-through modes in terms of their realism for user perception, particularly with volumetric avatars, warrants further investigation. Further, understanding how AR depth perception influences spatial behavior and movement patterns could provide valuable insights for system design. Expanding this research to include multi-person interactions and developing more advanced frameworks for movement analysis would further enrich this kind of studies and the field more generally.

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